

TRUSTSHIELD CLAIMS PROCESSING AI

Enterprise Workflow Blueprint

Role-Based Claims Triage, Review & Governance Console (Prototype)

Focus Area: Role-based authentication and navigation, AI-assisted claims triage across Auto / HITL / Exception pathways, case decision replay, auditor compliance review, and administrator controls for users, autonomy thresholds, and model release gates.

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Table of Contents

1. Purpose of This Document	
2. Platform Overview — Login Experience	
3. Authentication & Role Routing	Fig. 1
4. Persona Access Matrix (Sidebar Navigation by Role)	Fig. 2
5. Persona Workflows	
5.1 Operational Dashboard & Case Triage	Fig. 3
5.2 Case Decision Replay Page	Fig. 4
5.3 Auditor Console	Fig. 5
5.4 Admin Console	Fig. 6
5.5 Executive Dashboard	Fig. 7
6. App Shell & Layout Composition	Fig. 8
7. Data Model Summary	
8. Build Decisions Log	
9. Appendices	

1. Purpose of This Document

TrustShield is a role-based claims processing console. Four personas — **Operational User**, **Auditor**, **Executive Sponsor**, and **Admin** — authenticate against a single login screen and are routed to isolated workspaces scoped to their responsibilities. This document captures the current state of the prototype: how login and role resolution work, what each persona's screen contains, and how case data flows from intake through decisioning.

This blueprint reflects the actual implemented behaviour of the codebase at [ClaimsProcessingAI/src](#) as of this build cycle, not a proposed design — every diagram and table below traces to a specific component or data file in the repository.

Current Build Scope: Login & role routing, role-scoped sidebar navigation, Operational Dashboard (case triage across Auto / HITL / Exception categories), Case Decision Replay page, Auditor Console (decisioned-case review), Admin Console (Users & Roles, Autonomy Envelope, Release Gate), and Executive Dashboard (POC goal cards, hours-saved and STP metrics, outcome chart, recent decisions). All five persona-facing routes are now implemented — no placeholder screens remain.

2. Platform Overview — Login Experience

The public entry point (`src/pages/Login.tsx`) is a single sign-in screen with an email and password form, plus a “Demo roles” picker that lists all four personas from `src/data/roles.json` . Selecting a demo role auto-fills its credentials so reviewers can switch personas without memorising passwords; submitting still runs the same credential check as a real sign-in.

- Credentials are matched against `roles.json` by exact email + password.
- On success, `setAuthedUser()` persists `{ id, label, personName, email, route }` to `localStorage` under the key `trustshield:auth` .
- The user is immediately navigated to that role's fixed landing route.

Demo role rows show only the role label (channel tags and “Preview” badges were removed per build feedback — every listed role now behaves identically at sign-in).

3. Authentication & Role Routing

Role resolution happens entirely client-side against the static `roles.json` credential list. There is no separate token-exchange step in this prototype — the matched role record's `route` field is the single source of truth for where a persona lands.

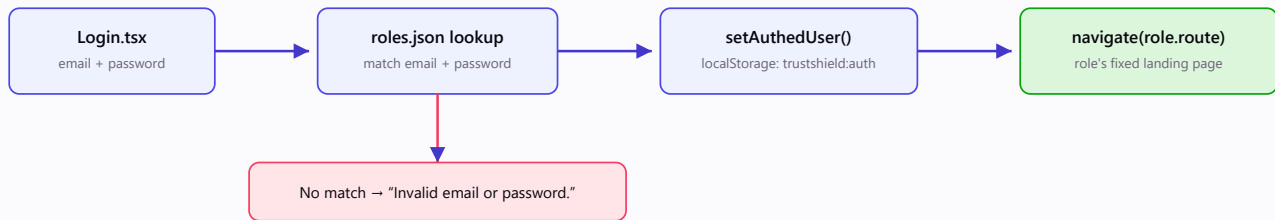


Fig. 1 — Authentication & role routing flow (Login.tsx → auth.ts → role landing route)

Role ID	Label	Landing Route	Console Status
operational	Operational User	<code>/dashboard</code>	Implemented
auditor	Auditor	<code>/auditor</code>	Implemented
executive	Executive Sponsor	<code>/executive</code>	Implemented
admin	Admin	<code>/admin</code>	Implemented

On every subsequent route render, `AppLayout.tsx` reads `getAuthedUser()` ; if no user is stored it redirects to `/login` via `<Navigate>` , so the session persists across reloads until `clearAuthedUser()` runs on sign-out.

4. Persona Access Matrix — Sidebar Navigation by Role

`AppLayout.tsx` passes the signed-in user's `roleId` into `Sidebar.tsx`, which looks the role up in a fixed `ROLE_NAV_ITEMS` dictionary. Every role always sees a “Home” link, plus exactly the nav items mapped to its own role key — there is no shared master list being filtered, so a role can only ever render its own entries.

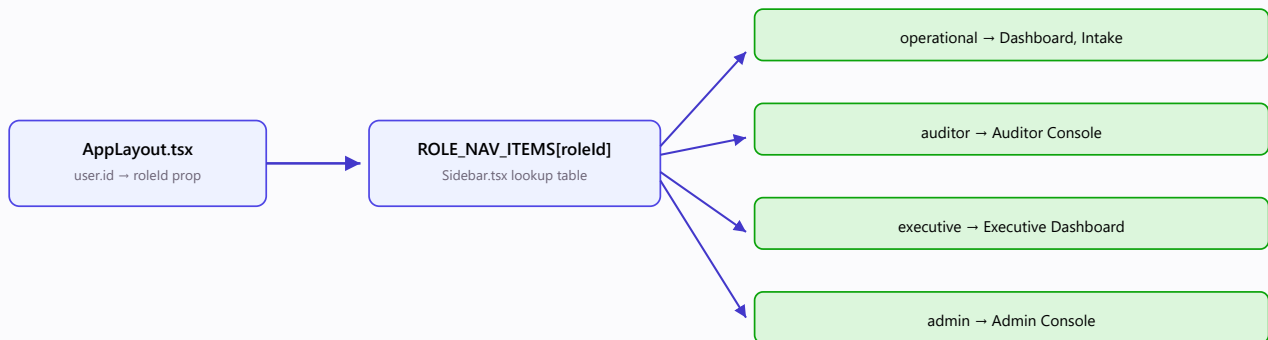


Fig. 2 — Sidebar navigation resolved per role via `ROLE_NAV_ITEMS` lookup (`Sidebar.tsx`)

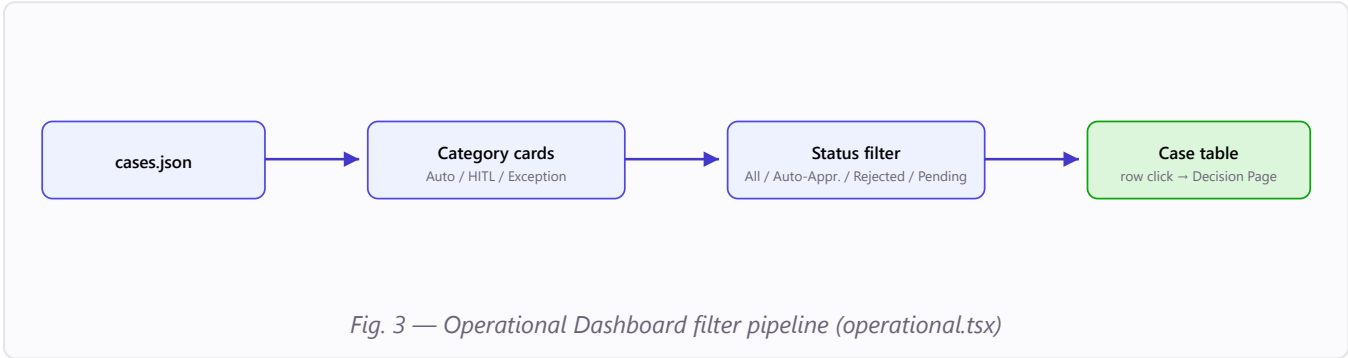
Persona	Sidebar Items	Primary Screen Purpose	Route Guard
Operational User	Home, Dashboard, Intake	Triage Auto / HITL / Exception cases, approve or reject pending items	AppLayout redirects to /login if unauthenticated
Auditor	Home, Auditor Console	Review decisioned (non-Pending) cases: Approved / Auto-Approved / Rejected	Same as above
Executive Sponsor	Home, Executive Dashboard	POC goal tracking, hours-saved & STP metrics, outcome chart, decision drill-down	Same as above
Admin	Home, Admin Console	Manage users, autonomy thresholds, model release gates	Same as above

Note: Route-level access is not yet persona-guarded beyond the login check — any signed-in user who types `/admin` or `/auditor` directly can reach that screen regardless of their own role. The sidebar link visibility is the only current access boundary.

5. Persona Workflows

5.1 Operational Dashboard & Case Triage

`src/pages/dashboard/operational.tsx` reads every case from `cases.json` and groups them into three routing categories: **Auto** (straight-through AI processing), **HITL** (human-in-the-loop review), and **Exception** (escalated for manual handling). Clicking a category card filters the table below it; a status dropdown (Auto-Approved / Approved / Rejected / Pending) layers on a second filter. Clicking any row navigates to that case's Decision Replay page.



Category	Meaning	Typical Status Outcome
Auto	Straight-through processed by AI, no human step	Auto-Approved
HITL	Human-in-the-loop review of an AI recommendation	Approved / Rejected / Pending
Exception	Escalated for manual handling (low AI confidence)	Pending / Approved / Rejected

5.2 Case Decision Replay Page

`src/pages/dashboard/decision.tsx` (route `/dashboard/cases/:caseId`) is the shared drill-down every persona's table row links to. It joins a case record from `cases.json` with its extended detail record in `decision.json` and renders:

- Case header — case ID, category badge, status badge, claimant name & city
- **Reviewer Decision** card — Approve / Reject buttons, shown only for non-auto, still-Pending cases; the choice is persisted to `localStorage` per case ID
- Case Details, Rule Evaluation (passed / flagged / failed), Submitted Documents (masked/reveal toggle), Case Timeline, and Decision Pattern & Reasoning

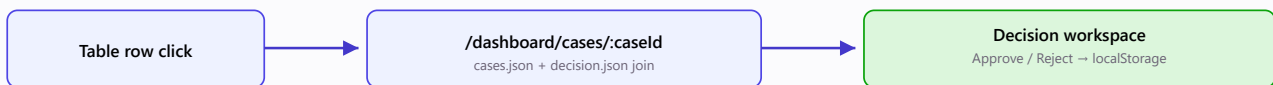
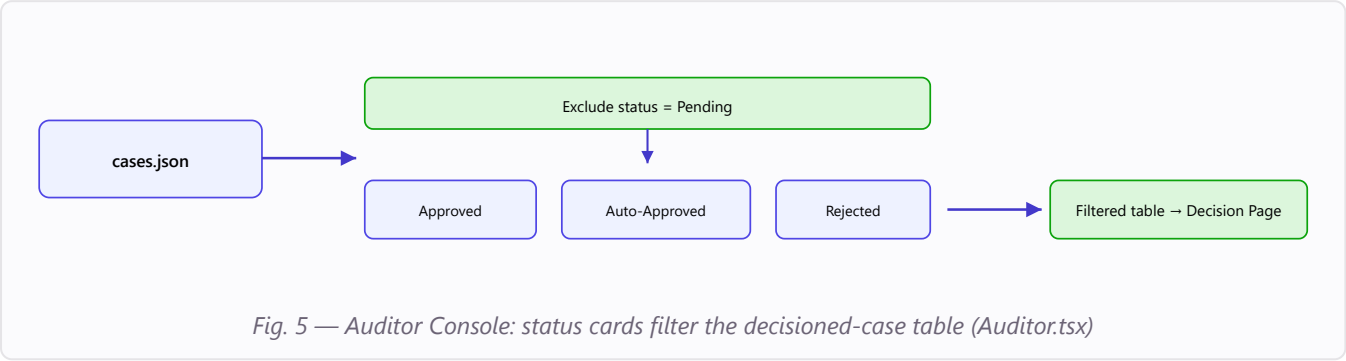


Fig. 4 — Decision Replay page composition (`decision.tsx`)

5.3 Auditor Console

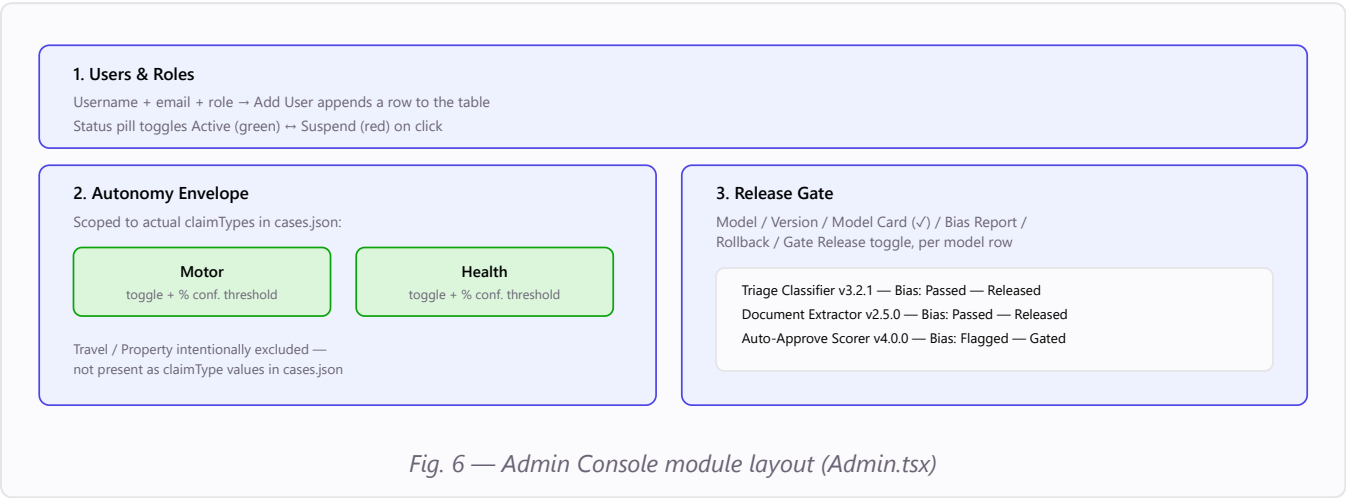
`src/pages/dashboard/Auditor.tsx` (route `/auditor`) is scoped to already-decided cases only. Pending cases are always excluded from view. Three summary cards — **Approved**, **Auto-Approved**, **Rejected** — each show a live count from `cases.json` ; clicking a card toggles it as a table filter (click again to clear). Row click navigates to the same Decision Replay page used by the Operational Dashboard.



Card	Filter Behaviour
Approved	Shows only Approved HITL/Exception cases
Auto-Approved	Shows only Auto-Approved straight-through cases
Rejected	Shows only Rejected cases
(none selected)	Shows all non-Pending cases combined

5.4 Admin Console

`src/pages/dashboard/Admin.tsx` (route `/admin`) is split into three modules, all backed by local component state (no persistence yet):

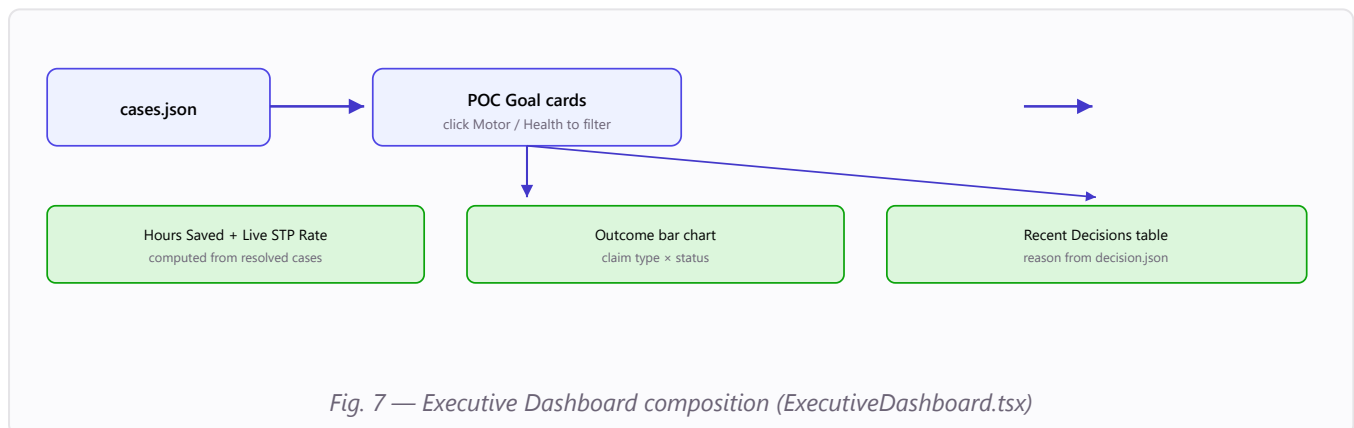


Module	Key Interactions
Users & Roles	Add user (username, email, role dropdown); toggle a user's status pill between Active / Suspend
Autonomy Envelope	Per claim class (Motor, Health) toggle auto-settlement eligibility and set a 0–100% AI-confidence threshold required to auto-approve without human review
Release Gate	Per model row: view model card (checkmark icon), view bias report status, rollback a release, toggle Released / Gated

5.5 Executive Dashboard

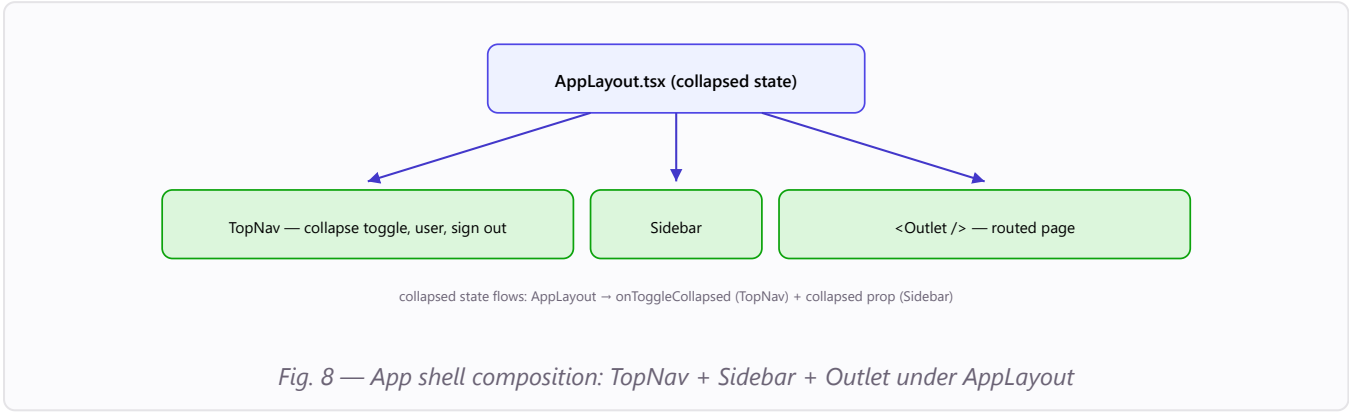
`src/pages/dashboard/ExecutiveDashboard.tsx` (route `/executive`) replaces the earlier placeholder screen. It ties the platform's live case data back to the POC-level business targets defined in the companion PRD/Approach documents (Section 3.2 / 15), rather than showing generic engagement metrics.

- **POC Goals row** — five cards for the Week-10 targets: Motor-claims TAT (12d→3d), Health-claims TAT (18d→7d), STP rate (14%→42%), Leakage reduction (–25%), Customer NPS (+15 pts). Clicking a claim-type goal (Motor / Health) filters the decisions table below it.
- **Total Adjudication Hours Saved** — Auto-Approved cases count as zero adjuster minutes; HITL/Exception cases use a 45-min-manual vs. 12-min-AI-assisted baseline to compute minutes saved, aggregated to hours.
- **Live STP Rate** — computed directly from `cases.json` as $\text{Auto-Approved} \div \text{all decisioned cases}$, shown against the 14%→42% target.
- **Decisions by Claim Type & Outcome** — a stacked bar chart (Motor / Health × Approved / Auto-Approved / Rejected), rendered with `recharts`.
- **Recently Completed Decisions table** — case ID, category, claim type, time since received, status, the reason pulled from `decision.json`, and an AI/manual time comparison; responds to the goal-card claim-type filter.



6. App Shell & Layout Composition

Every authenticated route renders inside `AppLayout.tsx`, which composes a fixed `TopNav` and role-scoped `Sidebar` around a routed `<Outlet />`. Sidebar collapse state is owned by `AppLayout` and lifted to both siblings.



Route table as currently registered in `App.tsx`:

Path	Element	Layout
/	Home	None (public)
/login	Login	None (public)
/dashboard	OperationalDashboard	AppLayout
/dashboard/cases/:caseId	DecisionPage	AppLayout
/intake	IntakePage	AppLayout
/auditor	Auditor	AppLayout
/executive	ExecutiveDashboard	AppLayout
/admin	Admin	AppLayout

7. Data Model Summary

All screens read from static JSON fixtures under `src/data/` — there is no backend API in this prototype.

File	Shape	Consumed By
<code>cases.json</code>	Array of cases: caseld, category (auto/hitl/exception), claimType (Motor/Health), status, aiConfidence, aiRecommendation, humanAction, overridden, fnolld, channel, receivedAt, photoAI, documentAI	operational.tsx, Auditor.tsx, decision.tsx, ExecutiveDashboard.tsx
<code>decision.json</code>	Record keyed by caseld: claimant, incident, rules[], documents[], timeline[], decision pattern & reasoning	decision.tsx, operational.tsx (claimant name lookup), ExecutiveDashboard.tsx (reason text)
<code>roles.json</code>	Array of role credentials: id, label, personName, channel, email, password, route, implemented	Login.tsx
<code>intake.json</code>	Intake-specific fixture data	intake.tsx

Client-side persisted state (all via `localStorage` , no server):

- `trustshield:auth` — the signed-in user object
- `trustshield:reviewer-decision:{caseId}` — per-case Approve/Reject choice made on the Decision Replay page

8. Build Decisions Log

Notable decisions made and course-corrections applied while building this prototype, kept here so the rationale isn't lost:

Area	Decision	Reason
index.css	Google Fonts <code>@import</code> reordered before <code>@import "tailwindcss"</code>	CSS spec requires <code>@import</code> to precede all other statements; Tailwind's import expands into real rules first otherwise
Auditor Console	Cards limited to Approved / Auto-Approved / Rejected; Pending always excluded from both cards and table	Auditor reviews decisioned cases only
Admin – toggle switches	Switched from absolute-positioned thumb to <code>inline-flex</code> + padding + <code>overflow-hidden</code>	Thumb circle was rendering outside the pill border
Admin – Autonomy Envelope	Removed Travel and Property claim classes; kept only Motor and Health	Those are the only <code>claimType</code> values that actually appear in <code>cases.json</code> — Travel/Property were fabricated
Admin – Autonomy Envelope	Added a per-class % confidence threshold input alongside the enable toggle	A toggle alone didn't express at what AI confidence auto-settlement is allowed
Admin – Release Gate	Model Card column changed from a "View" text button to a centered green checkmark icon	Matches requested visual reference
Login – Demo roles	Removed the <code>(channel)</code> label and the "Preview" badge from each role row	Simplify the picker to just the role label
Executive Dashboard	Built out <code>ExecutiveDashboard.tsx</code> to replace the <code>RolePlaceholder</code> screen at <code>/executive</code>	Referenced an existing Executive Dashboard from a sibling project, then re-grounded every metric in TrustShield's own PRD/Approach targets and actual <code>cases.json</code> / <code>decision.json</code> data instead of copying the reference verbatim
App.tsx	Removed the now-unused <code>RolePlaceholder</code> import	All four persona routes are implemented; nothing references it anymore

9. Appendices

Appendix A — Key Source Files

- `src/lib/auth.ts` — localStorage-backed auth helpers (getAuthedUser, setAuthedUser, clearAuthedUser)
- `src/components/layout/AppLayout.tsx` , `TopNav.tsx` , `Sidebar.tsx`
- `src/components/case-meta.ts` , `case-badges.tsx` — shared category/channel/status styling
- `src/pages/dashboard/operational.tsx` , `decision.tsx` , `Auditor.tsx` , `Admin.tsx` , `ExecutiveDashboard.tsx`
- `src/pages/Login.tsx` , `Home.tsx` , `intake.tsx`

Appendix B — Glossary

Term	Definition
Autonomy Envelope	The Admin-configured confidence threshold per claim class above which AI may auto-settle a claim without human review
HITL	Human-in-the-loop — a case category requiring an operator to confirm or override the AI recommendation
Decision Replay Page	The shared read/decide screen (<code>/dashboard/cases/:caseId</code>) every persona's table links to for a given case
Release Gate	Admin-controlled toggle per model version determining whether it is live (Released) or held back (Gated)

Appendix C — Known Gaps

- No server-side persistence — all writes (users, thresholds, decisions) live only in component state or localStorage and reset on reload where noted.
- No route-level RBAC guard beyond the login redirect — sidebar link visibility is the only current access control.
- Executive Dashboard's TAT/Leakage/NPS goal cards show static PRD targets rather than metrics computed live from data (only STP rate and hours-saved are computed live); the underlying data doesn't yet carry resolution timestamps needed to compute real TAT.